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IET ASSIGNMENT

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Course Code: DSA0402

Course Name: Fundamentals of Data Science

PROBLEM STATEMENT

Develop a machine learning-based system to predict whether bank customers will subscribe to a term deposit using customer and marketing campaign data, and segment customers into meaningful groups using K-Means clustering for better targeted marketing decisions.

1. Introduction

Banks regularly conduct marketing campaigns to encourage customers to subscribe to financial products such as term deposits. However, customers do not respond to these campaigns in the same way. Understanding customer characteristics and predicting their response can help banks make their marketing activities more focused.

This project develops a machine learning-based system that predicts whether a bank customer is likely to subscribe to a term deposit. Along with prediction, K-Means clustering is used to group customers with similar characteristics. The project therefore combines both customer subscription prediction and customer segmentation in a single data science workflow.

2. Problem Statement

The main problem addressed in this project is to predict the subscription behaviour of bank customers based on their demographic, financial and campaign-related information.

The project also aims to identify groups of customers having similar characteristics so that different customer segments can be understood and targeted more effectively.

3. Objectives

The main objectives of the project are:

- To load and analyze the Bank Marketing dataset.
- To preprocess numerical and categorical customer information.
- To perform exploratory data analysis using graphs.
- To calculate descriptive statistical measures.
- To calculate a 95% confidence interval.
- To build kNN, Decision Tree and Logistic Regression models.
- To compare the performance of the classification models.
- To predict the subscription behaviour of a new customer.
- To perform customer segmentation using K-Means clustering.
- To interpret the obtained customer segments.

4. Dataset Description

The project uses the Bank Marketing dataset obtained from the UCI Machine Learning Repository.

The dataset contains information collected from direct marketing campaigns of a Portuguese banking institution. The bank-full.csv file used in this project contains 45,211 customer records and 17 attributes.

The dataset contains both customer-related and campaign-related information. The target variable is y, which indicates whether the customer subscribed to a term deposit. The target contains two classes: yes and no.

Some important attributes in the dataset are:

Attribute	Description
age	Age of the customer
job	Occupation
marital	Marital status
education	Education level
default	Credit default status
balance	Average yearly balance
housing	Housing loan status
loan	Personal loan status
contact	Contact communication type
day	Day of the month
month	Month of contact
duration	Duration of the contact
campaign	Number of contacts during the campaign
pdays	Number of days since previous contact
previous	Number of previous contacts
poutcome	Outcome of previous campaign
y	Term-deposit subscription

5. Tools and Technologies

The project was implemented using Python.

The major libraries used were:

- **Pandas** for data loading and manipulation.
- **NumPy** for numerical operations.

- **SciPy** for statistical calculations.
- **Matplotlib** for creating graphs.
- **Seaborn** for statistical visualization.
- **Scikit-learn** for preprocessing, machine learning and clustering.

The complete implementation is contained in the main.py file.

6. System Workflow

The project follows a complete data science workflow starting from dataset loading and ending with customer segmentation.

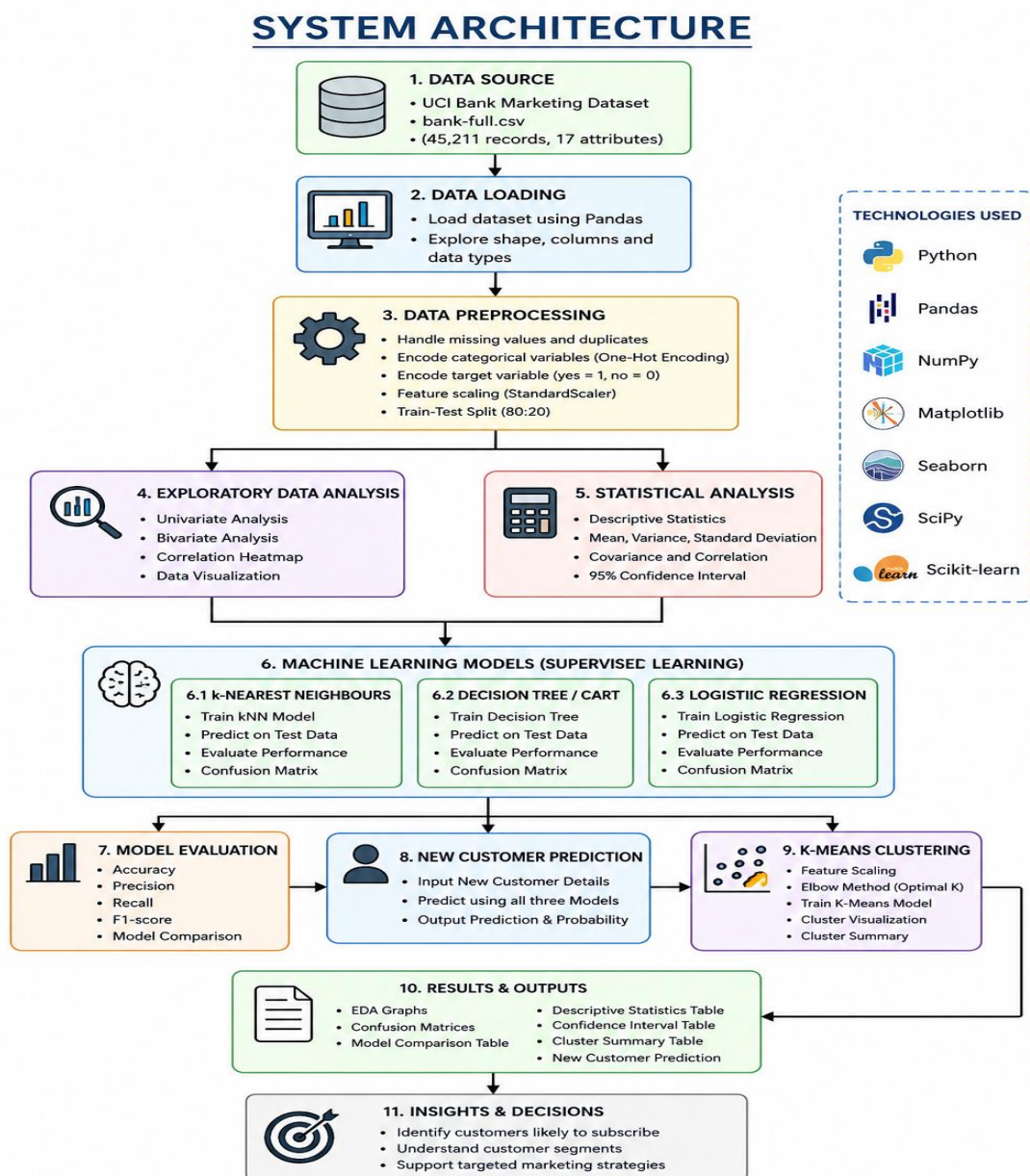


Figure 1: Overall workflow of the proposed system

7. Data Preprocessing

The dataset was loaded using Pandas and its structure was examined before performing the analysis. The dataset contains numerical as well as categorical attributes.

The data was checked for missing values and duplicate records. Categorical variables were converted into numerical representations so that they could be used by machine learning algorithms.

The target variable y was encoded into a binary form for classification. Numerical features were standardized where required. Scaling is particularly important for kNN and K-Means because both methods depend on distance calculations.

The processed data was then divided into training and testing sets for the supervised learning models.

8. Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the distribution of customer information and identify relationships between important variables.

The generated visualizations include customer age distribution, age versus subscription behaviour and correlation between numerical attributes.

8.1 Age Distribution

The age distribution was plotted to understand the age profile of customers in the dataset.

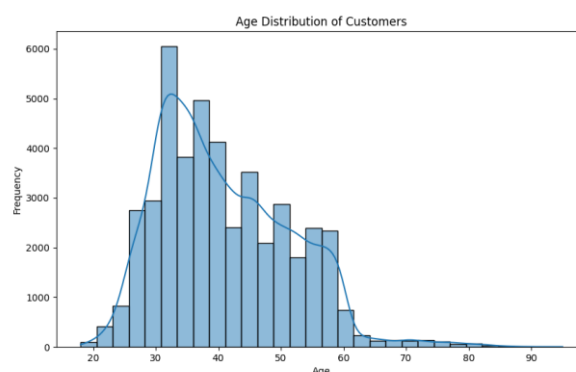


Figure 2: Age distribution of customers

The graph shows that most customers are concentrated within the middle-age range, while relatively fewer customers are present at the extreme age ranges.

8.2 Age and Subscription

The relationship between customer age and subscription behaviour was also visualized.

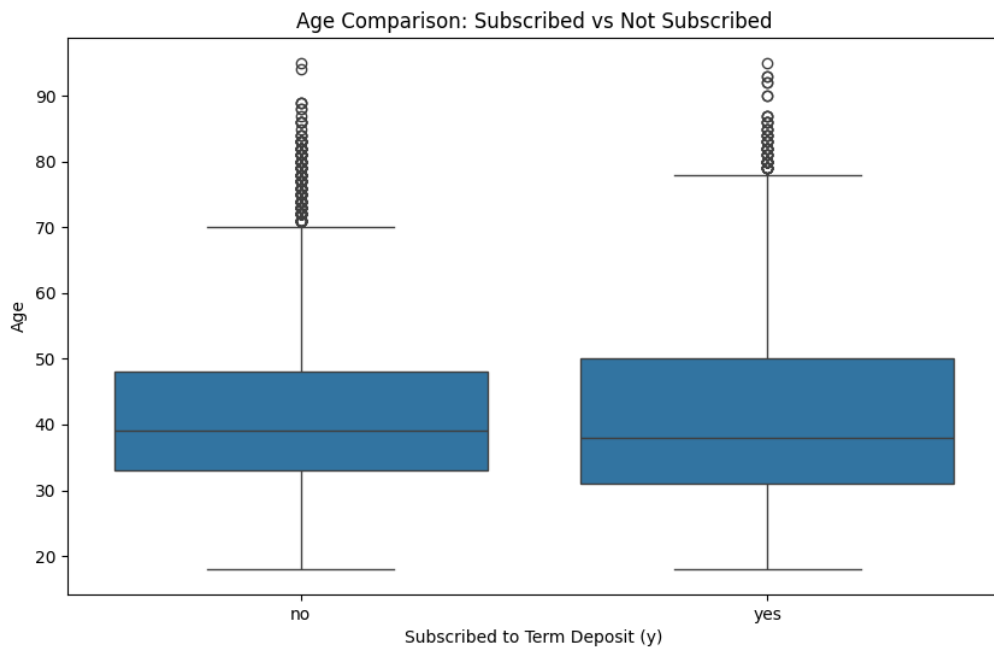


Figure 3: Relationship between customer age and subscription

This visualization helps compare subscription behaviour across different customer ages. It provides an initial indication that subscription behaviour is not identical across all age groups.

8.3 Correlation Analysis

A correlation heatmap was generated for the numerical variables in the dataset.

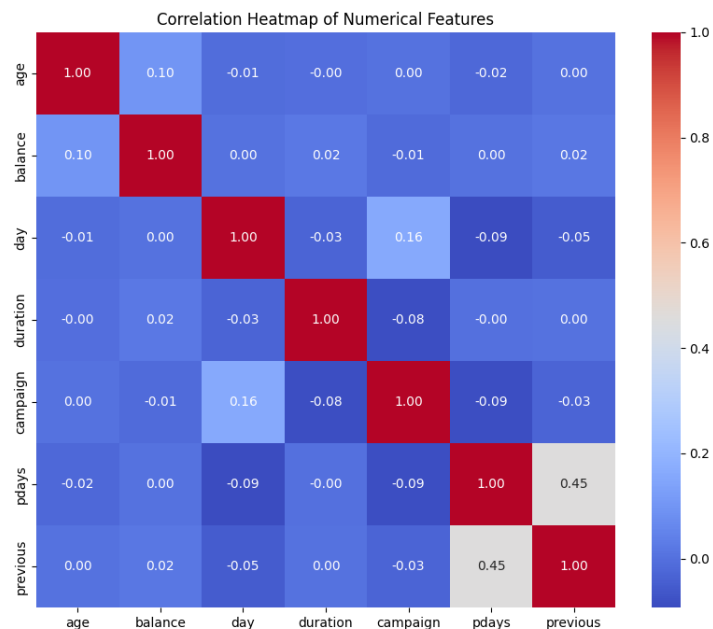


Figure 4: Correlation heatmap of numerical attributes

The heatmap provides an overview of the relationships between the numerical variables. Most variables do not show a very strong linear relationship with each other, although some variables have noticeable associations.

9. Descriptive Statistical Analysis

Descriptive statistics were calculated for the main numerical variables. The analysis included count, mean, standard deviation, minimum, quartiles and maximum values.

The dataset contains **45,211 observations** for each of the selected numerical variables.

Some important calculated values are shown below:

Statistic	Age	Balance	Day	Duration	Campaign	Pdays	Previous
Mean	40.9362	1362.2721	15.8064	258.1631	2.7638	-40.645? *	0.552?*
Minimum	18	-8019	1	0	1	-1	0
Median	39	448	16	180	2	-1	0
Maximum	95	102127	31	4918	63	871	275

Table 1: Descriptive statistics of selected numerical attributes

Note: For the pdays and previous mean values, copy the exact values directly from your descriptive_statistics.csv rather than using the shortened values above.

The results show considerable variation in some customer attributes. For example, customer balance ranges from a negative value to a very high positive value. Contact duration also has a wide range, indicating that campaign interactions can differ substantially between customers.

10. Confidence Interval

A 95% confidence interval was calculated as part of the statistical analysis to estimate the population mean of the selected customer variable.

The confidence interval uses the sample mean and standard error to estimate a range of plausible values for the corresponding population mean.

Table 2: 95% confidence interval

Measure	Value
Sample Size (n)	45,211

Sample Mean Age	40.9362 years
Standard Deviation	10.6188 years
Standard Error	0.0499
Confidence Level	95%
Lower Confidence Limit	40.8383 years
Upper Confidence Limit	41.0341 years

The confidence interval provides an estimate of where the population mean is expected to lie at the 95% confidence level. The width of the interval represents the uncertainty associated with estimating the population parameter from the available data.

11. Supervised Machine Learning

Three classification algorithms were implemented for predicting customer subscription:

1. k-Nearest Neighbours (kNN)
2. Decision Tree / CART
3. Logistic Regression

The dataset was divided into 80% training data and 20% testing data. The models were trained using the training data and evaluated using the unseen testing data.

11.1 k-Nearest Neighbours

The k-Nearest Neighbours algorithm predicts the class of a new observation based on the classes of its nearest neighbouring observations.

Feature scaling was applied before using kNN because the algorithm depends on distance calculations. The model was trained using the prepared training data and then used to predict the subscription status of test customers.

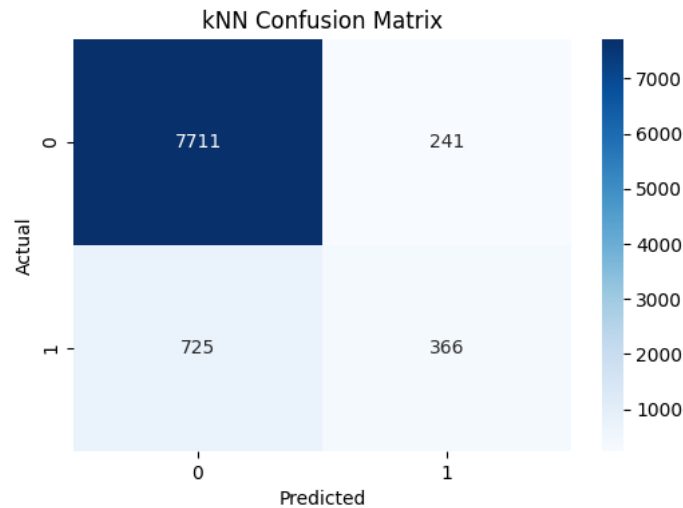


Figure 5: Confusion matrix of the kNN model

The confusion matrix shows the number of correctly and incorrectly classified customers for the two subscription classes.

11.2 Decision Tree / CART

The Decision Tree model classifies customers by creating a series of decision rules based on the input features. CART was used to construct the tree-based classification model.

The model was trained on the training dataset and evaluated using the test dataset.

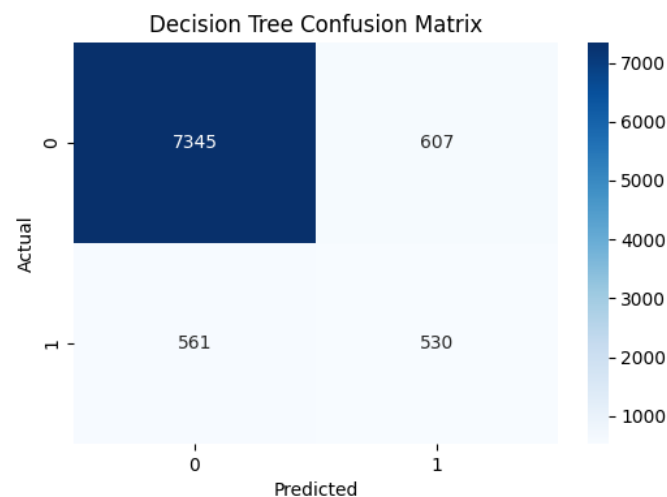


Figure 6: Confusion matrix of the Decision Tree model

The confusion matrix provides a clear view of the correct and incorrect predictions made by the Decision Tree model.

11.3 Logistic Regression

Logistic Regression was used to estimate the probability that a customer would subscribe to the term deposit.

The model uses the input attributes to calculate the probability of belonging to the subscription class and then produces the final classification.

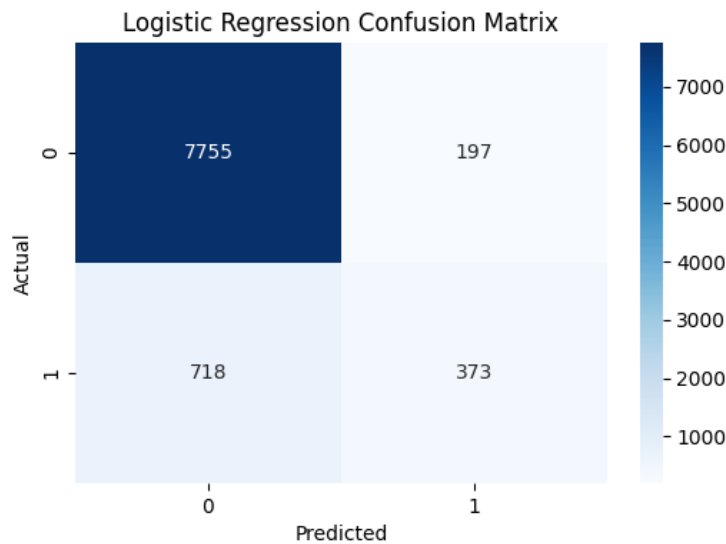


Figure 7: Confusion matrix of the Logistic Regression model

The confusion matrix shows the classification performance of the Logistic Regression model on the test data.

12. Model Evaluation and Comparison

The models were evaluated using **Accuracy, Precision, Recall and F1-score**.

The actual results obtained from model_comparison.csv are:

Model	Accuracy	Precision	Recall	F1-score
kNN	0.8932	0.6030	0.3355	0.4311
Decision Tree	0.8708	0.4661	0.4858	0.4758
Logistic Regression	0.8988	0.6544	0.3419	0.4491

Table 3: Performance comparison of classification models

From the results, Logistic Regression achieved the highest accuracy of approximately 89.88% and the highest precision of approximately 65.44%.

The Decision Tree achieved the highest recall at approximately 48.58% and the highest F1-score at approximately 47.58% among the three models.

Therefore, the choice of the best model depends on the objective. Logistic Regression is stronger in terms of overall accuracy and precision, while the Decision Tree performs better in identifying a larger proportion of actual subscribers and provides the highest F1-score.

13. New Customer Prediction

The trained models were also used to simulate the prediction for a new customer.

The program produced the following results:

Model	Prediction
kNN	No
Decision Tree	No
Logistic Regression	No

Table 4: New customer subscription prediction

All three models predicted that the simulated customer would not subscribe to the term deposit.

The Logistic Regression model produced a subscription probability of approximately 0.0081, which is a very low probability for the positive subscription class.

This demonstrates how the trained models can be used to estimate the likely response of an individual customer.

14. K-Means Customer Segmentation

K-Means clustering was used to divide customers into groups with similar characteristics.

The selected numerical features were scaled before clustering so that features with larger numerical ranges would not dominate the distance calculation.

The Elbow Method was used to examine different values of K and select an appropriate number of clusters.

14.1 Elbow Method

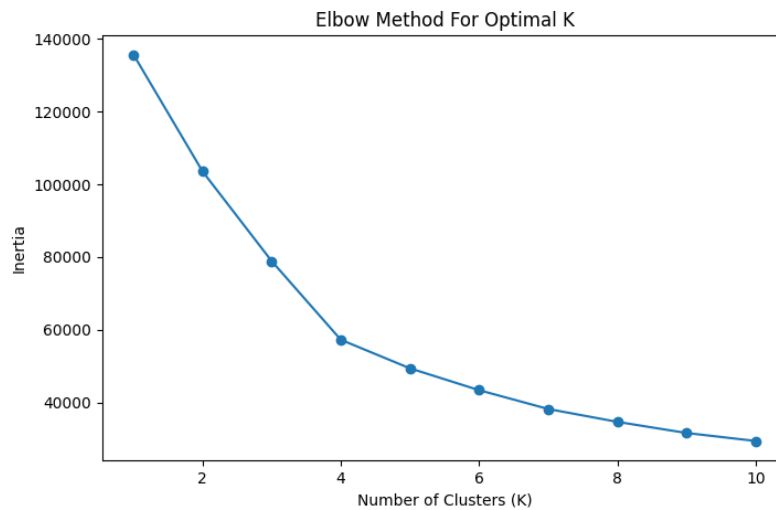


Figure 8: Elbow Method for determining the number of clusters

The Elbow Method helps identify a suitable cluster count by observing the point where increasing the number of clusters provides progressively smaller improvements.

14.2 Cluster Visualization

After applying K-Means, the resulting clusters were visualized to understand the separation between the customer groups.

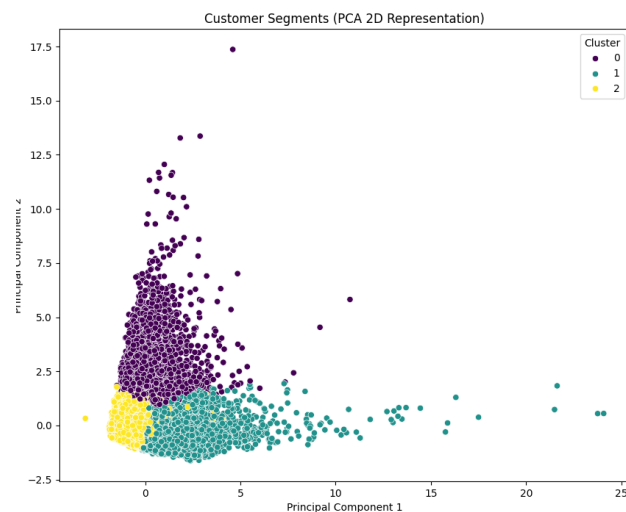


Figure 9: Visualization of customer clusters

The visualization shows the grouping of customers into three distinct clusters generated by the K-Means algorithm.

15. Customer Segment Analysis

The generated clusters were summarized using average age, balance, contact duration, subscription encoding and the number of customers in each cluster.

The actual cluster results are:

Cluster	Average Age	Average Balance	Average Duration	Subscription Rate	Customer Count
0	39.79	1450.11	873.50	0.4508	4,270
1	52.37	2051.15	195.24	0.0919	15,536
2	34.13	926.24	193.22	0.0762	25,405

Table 5: Summary of customer segments

The clustering results show clear differences between the three groups.

Cluster 0 contains 4,270 customers and has an average age of about 39.79 years. Its average balance is approximately 1450.11 and its subscription rate is about 45.08%, which is considerably higher than the other two clusters.

Cluster 1 contains 15,536 customers and has the highest average age and average balance among the three groups. However, its subscription rate is only about 9.19%.

Cluster 2 is the largest group, containing 25,405 customers. It has the lowest average age and average balance among the clusters and a subscription rate of about 7.62%.

These differences show that customer characteristics can vary considerably between segments.

16. Results and Discussion

The project successfully implemented an end-to-end data science workflow using the Bank Marketing dataset containing 45,211 customer records.

The exploratory analysis provided an understanding of the customer age distribution, subscription behaviour and relationships between numerical variables. The descriptive statistics also showed considerable variation in attributes such as balance and contact duration.

Three supervised learning algorithms were implemented for subscription prediction. The results show that Logistic Regression achieved the highest overall accuracy at approximately 89.88%, while kNN achieved approximately 89.32%. The Decision Tree achieved approximately 87.08% accuracy.

Although Logistic Regression had the highest accuracy and precision, the Decision Tree achieved the highest recall and F1-score. This shows why several evaluation metrics are useful when comparing classification models.

The new customer simulation produced a consistent No prediction from all three models, with Logistic Regression assigning a probability of approximately 0.0081 to subscription.

K-Means clustering divided the customers into three groups with noticeably different characteristics. The first cluster had a much higher subscription rate than the other two clusters, while the third cluster represented the largest portion of the customer base.

Overall, the results demonstrate that combining classification and clustering can provide both prediction of individual customer behaviour and a broader understanding of customer groups.

17. Conclusion

This project successfully developed a machine learning-based system for bank customer subscription prediction and customer segmentation.

The Bank Marketing dataset was first processed and explored using Python. Statistical analysis and confidence interval estimation were then performed to understand the numerical characteristics of the data.

Three machine learning algorithms, namely kNN, Decision Tree/CART and Logistic Regression, were implemented for subscription prediction. Their performance was compared using Accuracy, Precision, Recall and F1-score.

K-Means clustering was also applied to segment customers into groups with similar characteristics. The cluster analysis revealed differences in age, balance, contact duration and subscription rate between the customer groups.

The final implementation successfully performs the complete workflow automatically and generates the required graphs, tables, predictions and clustering results.

18. Future Enhancements

The current system can be improved in several ways:

- Hyperparameter tuning can be applied to improve model performance.
- Additional classification algorithms can be tested.
- Feature selection can be performed to identify the most useful attributes.
- Different clustering techniques can be compared with K-Means.
- A web-based interface can be developed for entering customer details and obtaining predictions.
- Updated banking campaign data can be incorporated for continuous analysis.
- The customer segments can be combined with campaign history to create more targeted marketing strategies.

19. References

1. UCI Machine Learning Repository – Bank Marketing Dataset.
2. Pandas Documentation.
3. NumPy Documentation.
4. SciPy Documentation.
5. Matplotlib Documentation.
6. Seaborn Documentation.
7. Scikit-learn Documentation.